El problema consiste en una pared compuesta por acero inoxidable y un material no metálico, expuesta a convección con vapor en la superficie interior y aire en la superficie superior

Modificaciones en la adaptación:

1) conductividad térmica $(k_2) = 3 \frac{w}{m \cdot K} \rightarrow 12 \frac{w}{m \cdot K}$

2) $(h_2) = 50 \frac{w}{m^2 \cdot K} \rightarrow 150 \frac{w}{m^2 K}$

Datos:

| | | | |
|---|---|---|---|
| $-k_1$ | $13 \frac{w}{m \cdot K}$ | $-L_2$ | $0.010\ m$ |
| $-k_2$ | $12 \frac{w}{m \cdot K}$ | $-h_1$ | $300 \frac{w}{m^2 K}$ |
| $-L_1$ | $0.030\ m$ | $-h_2$ | $150 \frac{w}{m^2 K}$ |
| $-T_{\infty,1}$ | $260°C$ | $-T_{\infty,2}$ | $20°C$ |
| $-\Delta x$ | $0.005\ m$ | | |

Assumptions:

1) Steady State
2) Conducción Unidimensional
3) Propiedades constantes
4) No égen

⇒ Resistencia Total

$$R''_{Total} = \frac{1}{h_1} + \frac{L_1}{K_1} + \frac{L_2}{K_2} + \frac{1}{h_2}$$

$$= \frac{1}{300} + \frac{0.030}{13} + \frac{0.010}{12} + \frac{1}{150} = \boxed{0.01314103 \ \frac{m^2 \cdot w}{K}}$$

$$R''_{Claude} = 0.013141 \ \frac{m^2 K}{w}$$

* Coinciden por redondeo

⇒ Flujo de calor

$$q'' = \frac{T_{\infty,1} - T_{\infty,2}}{R''_{Total}} = \frac{260 - 26}{0.01314103} = \boxed{18{,}263.41 \ w/m^2}$$

$$q''_{Claude} = \boxed{18{,}263 \ w/m^2}$$

* Coinciden por redondeo

→ Temperatura del nodo 0

$$q'' = h_1 (T_{\infty,1} - T_0) \rightarrow T_0 = T_{\infty,1} - \frac{q''}{h_1}$$

$$T_0 = 260 - \frac{18,263.41}{300} = \boxed{199.1220\ °C}$$

$T_{0\ claude} = 199.12\ °C$ * coinciden por redondeo

→ Verificar nodo 1

$$q'' = -k_1 \left( \frac{T_1 - T_0}{\Delta x} \right) \rightarrow T_1 = T_0 - \left( \frac{q'' \Delta x}{k_1} \right)$$

$$T_1 = 199.1220 - \frac{(18,263.41)(0.005)}{13} = \boxed{192.0976\ °C}$$

$T_{1\ claude} = \boxed{192.10\ °C}$ * coinciden por redondeo

→ Nodos 2-5

* ec. de diferencias finitas ⇒ $T_{i-1} - 2T_i + T_{i+1} = 0$

Nodo 2:

$$T_0 - 2T_1 + T_2 = 0 \Rightarrow T_2 = 2T_1 - T_0 = 2(192.0976) - 199.1220$$

$$\boxed{T_2 = 185.0732\ °C}$$

$T_{2\ Claude} = \boxed{185.07\ °C}$

Nodo 3: $T_1 - 2T_2 + T_3 = 0$

$T_3 = 2T_2 - T_1$

$T_3 = 2(185.0732) - 192.0976 = \boxed{178.0488\ °C}$

$T_{3\ Claude} = \boxed{178.05\ °C}$

Nodo 4: $T_2 - 2T_3 + T_4 = 0$

$T_4 = 2T_3 - T_2 = 2(178.0488) - 185.0732$

$\boxed{T_4 = 171.0244\ °C}$

$T_{4\ Claude} = \boxed{171.02\ °C}$

Nodo 5: $T_3 - 2T_4 + T_5 = 0$

$T_5 = 2T_4 - T_3 = 2(171.0244) - 178.0488$

$\boxed{T_5 = 164.00\ °C}$

$T_{5\ Claude} = \boxed{164.00\ °C}$

-> Nodo 6: Interfaz entre materiales

$$q'' = -k_1 \frac{T_6 - T_5}{\Delta x}$$

$$T_6 = T_5 - \frac{q'' \Delta x}{k_1} = 164.00 - \frac{18263.41\,(0.005)}{13}$$

$$T_6 = 156.9756\,°C$$

$$T_{6\,claude} = 156.98\,°C$$

-> Nodo 7

$$q'' = -k_2 \frac{T_7 - T_6}{\Delta x}$$

$$T_7 = T_6 - \frac{q'' \Delta x}{k_2} = 156.9756 - \frac{(18263.41)(0.005)}{12}$$

$$T_7 = 149.3659\,°C$$

$$T_{7\,claude} = 149.37\,°C$$

-> Balance de energía

$$\frac{k_1}{\Delta x}(T_5 - T_6) + \frac{k_2}{\Delta x}(T_7 - T_6) = 0$$

$$\frac{13}{0.005}(T_5 - T_6) + \frac{12}{0.005}(T_7 - T_6) = 0$$

$$2600\,(T_5 - T_6) + 2400\,(T_7 - T_6) = 0$$

$$2600\,T_5 - 2600\,T_6 + 2400\,T_7 - 2400\,T_6 = 0$$

$$2600\,T_5 - 5000\,T_6 + 2400\,T_7 = 0$$

$$\frac{}{200}$$

$$13\,T_5 - 25\,T_6 + 12\,T_7 = 0$$

$$13(164.00) - 25(156.9756) + 12\,(149.3659)$$

$$\text{Residuo} = 0.0008 \approx 0$$

→ Nodo 8:

$$q'' = h_2\,(T_8 - T_{\infty,2})$$

$$T_8 = T_{\infty,2} + \frac{q''}{h_2}$$

$$T_8 = 20 + \frac{18263.41}{150} = \boxed{141.7561\,°C}$$

$T_8$ clavado $= \boxed{141.76\,°C}$

→ Verificación de Boundary Condition superior

$$\frac{k_2}{\Delta x}\,(T_7 - T_8) + h_2\,(T_{\infty,2} - T_8) = 0$$

$$2400(T_7 - T_8) + 150(20 - T_8) = 0$$

$$2400T_7 - 2400T_8 + 3000 - 150T_8 = 0$$

$$\frac{2400T_7 - 2550T_8 + 3000 = 0}{150}$$

$$-T_7 + 1.0625T_8 = 1.25$$

$$-(149.3659) + 1.0625(141.7561) = 1.25$$

⇒ Verificar sistema matricial

| Fila | Ecuación |
|---|---|
| 0 | $1.115385T_0 - T_1 = 30$ |
| 1 | $-T_0 + 2T_1 - T_2 = 0$ |
| 2 | $-T_1 + 2T_2 - T_3 = 0$ |
| 3 | $-T_2 + 2T_3 - T_4 = 0$ |
| 4 | $-T_3 + 2T_4 - T_5 = 0$ |
| 5 | $-T_4 + 2T_5 - T_6 = 0$ |
| 6 | $13T_5 - 25T_6 + 12T_7 = 0$ |
| 7 | $T_6 - 2T_7 + T_8 = 0$ |
| 8 | $-T_7 + 1.0625T_8 = 1.25$ |

→ Comparación flujo de calor original y adoptado

* Verificación del aumento porcentual en el flujo de calor

$$q''_{original} = 8283.19 \ W/m^2 \quad , \quad q''_{adoptado} = 18263.41 \ W/m^2$$

$$\%\ aumento = \frac{q''_{adoptado} - q''_{original}}{q''_{original}} = \frac{18263.41 - 8283.19}{8283.19} \times 100$$

$$\boxed{\%\ aumento = 120.49\ \%}$$